

Instruction : Out of total 5 assignments, you are free to pick any 3 assignments including at least one “Conceptual Assignments”.

Conceptual Assignments (Write Handwritten Notes. Scan and upload PDF.)

1: Word Embeddings – Theory and Analysis

- **Objective:** Explain the concept of word embeddings.
- **Tasks:**
 - Compare one-hot encoding with dense word embeddings.
 - Explain the differences between Word2Vec (CBOW vs. Skip-Gram), GloVe, and FastText.
 - Discuss semantic similarity in embedding spaces with examples.
 - Analyze the role of dimensionality in embedding representations.

2: Embedding Space Exploration

- **Objective:** Explore vector relationships.
- **Tasks:**
 - Given a list of word embeddings, perform:
 - Cosine similarity computations between words.
 - Vector arithmetic (e.g., $\text{king} - \text{man} + \text{woman} \approx \text{queen}$).
 - Visualize embeddings using PCA or t-SNE. (**Optional if you can only**)
 - Interpret the visualized clusters (**Optional if you can only**).

Programming Assignments (Code) :

3: Build a Word2Vec Model

- **Objective:** Implement Skip-Gram or CBOW using a small corpus.
- **Tools:** Python, NumPy, or PyTorch/TensorFlow.
- **Tasks:**
 - Preprocess text (tokenization, vocabulary indexing).
 - Implement training using negative sampling.
 - Evaluate with similarity tasks or analogies.

4: Use Pre-trained Embeddings

- Objective: Apply pre-trained embeddings in downstream tasks.
- Tasks:
 - Load GloVe or FastText embeddings.
 - Use embeddings as input for(***You can choose any one of below task.***):
 - Sentiment classification (IMDB/Twitter dataset).
 - Named Entity Recognition (CoNLL dataset).
 - Compare performance with/without embeddings.

5: Contextualized vs Static Embeddings

- Objective: Compare static (Word2Vec/GloVe) and contextual embeddings (BERT/ELMo).
- Tasks:
 - Use Hugging Face Transformers to extract contextual embeddings.
 - Analyze sentence similarity or word sense disambiguation.
 - Report differences in performance/interpretability.